

- Some non-systemic insecticides, however, have ability to move from one surface leaf to the other.
- They are called as **trans laminar insecticides**.
- E.g., **Malathion, Diazinon, Spinosad etc.**

❖ **Biological control**

- Management of a pest by means of another living organism that is encouraged and disseminated by man is called biological control.

❖ **Techniques in biological control:**

- Biological control practices involve three techniques.
 - ✓ Introduction,
 - ✓ Augmentation
 - ✓ Conservation.

❖ **Parasites:**

- **Primary parasite:** A parasite attacking an insect which itself is not a parasite (Beneficial to man.)
- **Secondary parasite:** A hyperparasite attacking a primary parasite (Harmful to man)
- **Tertiary parasite:** A hyperparasite attacking a secondary parasite (Beneficial to man)
- **Quaternary parasite:** A hyperparasite attacking tertiary parasite (Harmful to man)

❖ **Microbial control**

- Microbial control refers to the exploitation of disease-causing organism to reduce the population of insect pest below the damaging levels.
- **Steinhaus (1949)** Coined the term 'Microbial Control' when microbial organisms or other products (toxins) are employed by man for the control of pests on plants, animals or man.

1. Bacteria:

- *Bacillus thuringiensis* (B.t.) is important and is isolated from flour moth, *Ephestia kuhniella* by Berliner (1915)
- Bt known as a bacterial insecticide is now being using by farmers mostly on lepidopterous larvae.

- The entry of the bacteria is by ingestion of the bacteria, which infect the mid gut epithelia cells and enter the hemolymph to sporulate and cause septicemia.
- Available in different formulations (Trade names Thuriocide, Delfin, Bakthane, Biobit, Halt, Dipel etc.).

2. Viruses:

❖ NPV (**Borrellina virus**):

- Nuclear polyhedral virus has been isolated from the order Lepidoptera.
- Among these viruses Baculoviruses (Baculoviridae) are successful in IPM.
- The virus infected dead larvae hanging upside down from plant parts (Tree top disease).
- The cuticle becomes fragile, rupturing easily when touched, discharges liquefied body fluids.
- NPV multiplies in insect body wall, trachea, fat bodies and blood cells.
- The polyhedra are seen in nuclei.
- The polyhedral bodies enlarge in size destroying the host nuclei to get released into the insect body cavity.

3. Fungi:

- The fungal disease occurrence in insects is commonly called as mycosis. Most of the entomopathogenic fungi infect the host through the cuticle.
- Important species are, *Entomophthora*, *Metarhizium*, *Beauveria*, *Nomuraea* and *Verticillium*.
- **Eg:** *Entomophthora grylli* on grasshoppers; *Aspergillus flavus* on *Epilachna* beetles; *Spicaria* sp. on castor whitefly; *Metarhizium anisopliae* (Green muscardine) on Orthoptera, white muscardine, *Beauveria bassiana* on *Leptinotarsa decemlineata*.

4. Entomopathogenic nematodes (EPNs)

- Rhabditids (Rhabditidae) are found to have a symbiotic relationship with the bacteria, forming a disease complex.

- The nematode serves as a vector for the bacterium, which produces a septicemia (sporulation in blood; Milky disease) in the insect body.
- The EPNs *Steinernema* spp and *Heterorhabditis* spp from the families, Steinernematidae and Heterorhabditidae have the mutualistic association with bacteria *Photobacterium* and *Xenorhabdus* spp., respectively.
- The only stage that survives outside the host is the non – feeding 3rd stage Infective Juvenile (IJ).
- *Rodolia cardinalis* (Vadalia beetle) against cotton cushion scale (*Icerya purchasi*)
- *Trichogramma ostrinae* against European corn borer.
- *Encarsia formosa* against greenhouse white fly.
- The spotted lady beetle (*Coleomegilla maculata*) is also able to feed on the eggs and larvae of the Colorado potato beetle (*Leptinotarsa decemlineata*)
- *Phasmarhabditis hermaphrodita* is a microscopic nematode that kills slugs.
- The bug *Orius insidiosus* has been successfully used against the two-spotted spider mite and the western flower thrips (*Frankliniella occidentalis*)
- *Bacillus thuringiensis* is the most widely applied species of bacteria used for biological control, with at least sub-species used against **Lepidopteran (moth, butterfly).**
- The bacterium *Paenibacillus popilliae* which causes milky spore disease has been found useful in the control of Japanese beetle, killing the larvae.
- *Beauveria bassiana* is mass-produced and used to manage a wide variety of insect pests including whiteflies, thrips, aphids and weevils.